PENINSULA COLLEGE GEORGETOWN DK266-03(P)

UNIVERSITY OF PLYMOUTH

COMPUTING GROUP PROJECT

PRESENTED BY: NFCampus

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01 INTRODUCTION

NFCampus: A Digital Student Identification and Access System Using NFC Technology for Android

- Purpose of the project is to provide a backup smartphone-based authentication system to reduce access disruptions while maintaining security.
- The project focuses on a controlled prototype and excludes replacing official campus cards, supporting iOS devices, or full campus deployment.

02 PROBLEM & APPROACH

Campus's Reader

[Relies]

Campus's Card

Lost

Forgotten

Damaged

Smartphones act as secure NFC clients

Store legitimate card UID and send via APDU commands

Custom reader (ESP32 + PN532) validates access against a whitelist

Android Machine

Host CPU

NFC Controller

[Application Data]

Custom Reader

03 OBJECTIVE

To create a secure, smartphone-based authentication system that acts as a digital backup for college access cards.

Key objectives include developing an Android HCE app, implementing a custom NFC reader, ensuring reliable access authorization, and delivering a fully functional proof-of-concept by March 28, 2026.

In-Scope

- User Authentication Flows
- OCR Integration and Image Capture
- NFC Card Binding
- Functional Android HCE
- Custom Reader (ESP32 + PN532) Validating UIDs

Out-of-Scope

- Campus Card Cloning
- iOS Support
- Live Database Integration
- Fully Functional Reader System for Buildings

SYSTEM

DEMO

04 FUTURE ENHANCEMENTS

Migrate to Cloud:

- Transition from local hosting to Cloud-based Backend Deployment

Strict Hardware Device Binding:

- Automatically block access on unauthorized devices and requiring termination of original account to register on new device.

Dynamic OCR & Parsing:

- Replace static word filters with AI-powered Natural Language Processing (NLP).

Smart Relink Workflow:

- Create a single-step process re-authentication to safely overwrite existing NFC data with a new one.

Time-Limited QR Tokens:

- QR codes will generate encrypted, time-stamped tokens that automatically expire after set amount of time.

Improved visuals & Design:

- Further enhance the design of the applications